



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2006
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 85

- Attempt questions 1 – 6
- Hand up in 3 separate booklets clearly labelled Section A, Section B and Section C

Examiner: *A. Fuller*

SECTION A

Question 1 (13 marks)

(a) Convert 80° to radians in exact form. [1]

(b) Convert $\frac{17\pi}{12}$ radians to degrees. [1]

(c) Differentiate the following:

(i) $4x^2 + 5$

(ii) $(2x^3 - 5)^4$ [2]

(iii) $\frac{3x+1}{2x-1}$ [2]

(iv) $x\sqrt{1-2x}$ [2]

(d) Find a primitive of:

(i) $\frac{2}{x^4}$

(ii) $\sqrt{x}$ [1]

(e) Find $f''(2)$ if $f(x) = x^5$ [2]

Question 2 (15 marks)

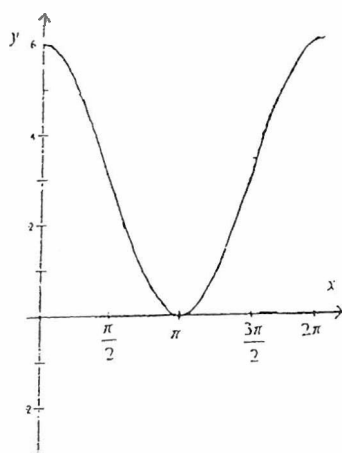
- (a) $A(-1,5)$, $B(2,1)$ and $C(4,k)$ are collinear. Find the value of k . [3]
- (b) Find $\int \frac{x^2+1}{x^2} dx$ [2]
- (c) Evaluate $\int_{-1}^1 (x-1)(x+1) dx$ [3]
- (d) A die is tossed twice. The sum of the numbers which appear on the upmost face of the die is calculated. Using a table or otherwise:
- (i) Find the probability that the sum is greater than 8. [2]
- (ii) It is known that a 4 appears on the die at least once in the two throws. Find the probability that the sum is greater than 8. [2]
- (e) The vertices of a triangle are $A(1,3)$, $B(8,2)$ and $C(4,-1)$.
- (i) Find the coordinates of D and E, the midpoints of AC and AB respectively.
- (ii) Hence, show that DE is parallel to CB. [2]

SECTION B

Question 3 (14 marks)

(a) If $y = (x^2 + 4)(x - 3)$, solve $\frac{dy}{dx} = 4$. [3]

(b) The diagram below is the graph of $y = 3 + 3 \cos x$



(i) Copy this graph onto your answer sheet.

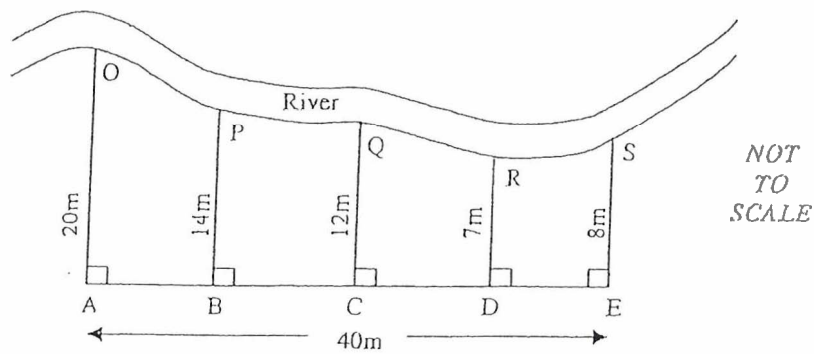
(ii) State the amplitude and period of $y = 3 \sin 2x$. [1]

(iii) On the same graph as (i), sketch $y = 3 \sin 2x$ for $0 \leq x \leq 2\pi$. [2]

(iv) How many solutions are there to the equation $3 + 3 \cos x = 3 \sin 2x$? [1]

(c) A class of 30 students contains 18 students who play cricket, 13 who play tennis and 5 who play both cricket and tennis. If one student is chosen at random find the probability that this student plays neither cricket nor tennis. [2]

- (d) The diagram below shows a paddock with one side bounded by a river. AE is a boundary fence 40 metres in length. AO, BP, CQ, DR, ES are offsets measured from the fence to the river with lengths as shown. $AB = BC = CD = DE$. [3]



Use Simpson's rule with 5 function values as shown on the diagram to approximate the area of the paddock.

- (e) The gradient function of a curve is $3x^2 - 1$ and the curve passes through the point $(4,1)$. Find the equation of the curve. [2]

Question 4 (13 marks)

- (a) A girl has 5 tickets in a raffle where 100 tickets are sold.
First prize is drawn discarded and then the second prize is drawn.
Find the probability that she wins:

(i) first prize [1]

(ii) second prize [2]

- (b) Consider the curve $y = x^4 - 4x^3$

(i) Find the coordinates of the stationary points. [2]

(ii) Determine the nature of these stationary points. [2]

(iii) For what values of x is the curve concave up? [1]

(iv) For what values of x is the curve decreasing? [1]

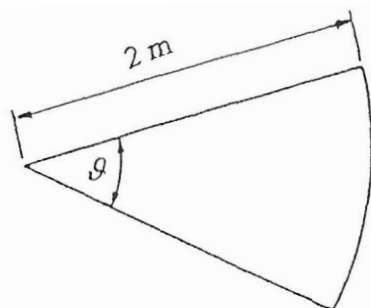
(v) Hence sketch the curve. [2]

- (c) If the probability that an event E occurs is $\frac{1}{x}$, express the [2]
probability that E does not occur as a single fraction.

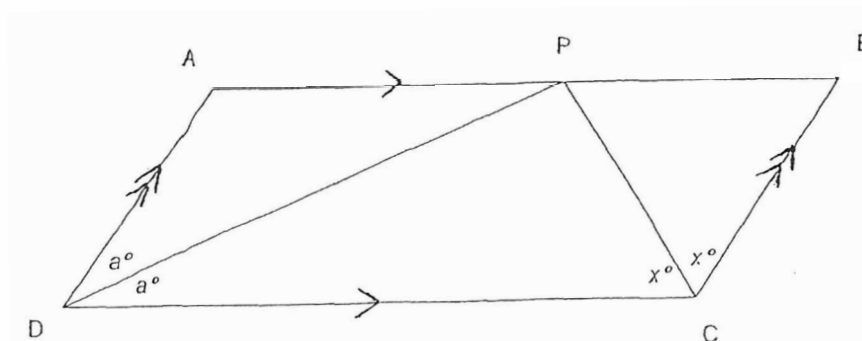
SECTION C

Question 5 (15 marks)

- (a) A flower bed is made in the shape of a minor sector with angle θ and radius 2 metres.

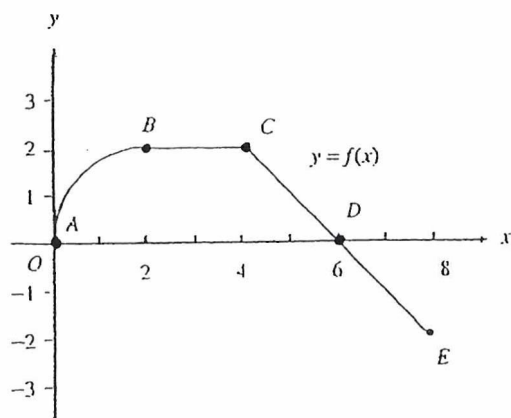


- (i) If the area of the flower bed is 1.6 m^2 , find the angle θ to the nearest minute. [2]
- (ii) Find the perimeter of the flower bed to the nearest cm. [2]
- (b) ABCD is a parallelogram. The bisectors of angles ADC and BCD meet at P on the side AB. Prove that:



- (i) $\angle DPC$ is a right angle. [2]
- (ii) $\triangle ADP$ is isosceles. [2]
- (iii) $AB = 2BC$ [2]

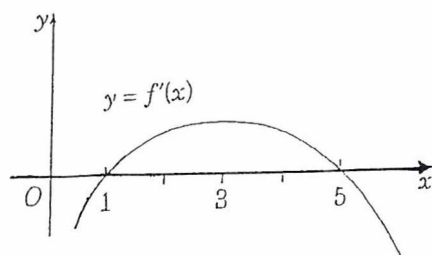
The graph below of the function f consists of a quarter circle AB and two line segments BC and CE.



(i) Evaluate $\int_0^8 f(x) dx$. [2]

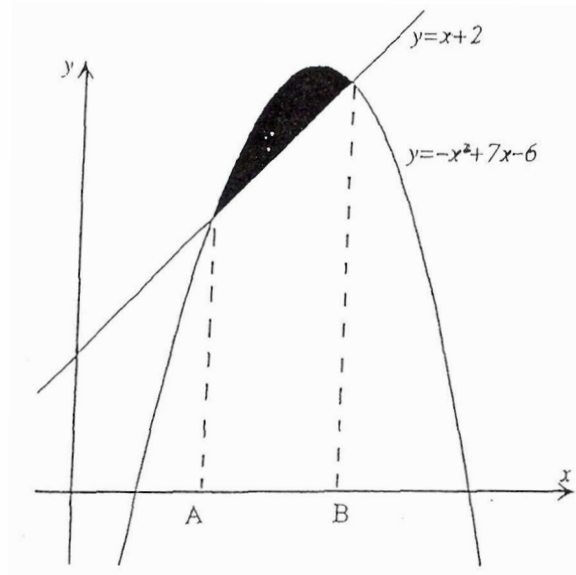
(i) For what value(s) of x satisfying $0 < x < 8$ is the function f not differentiable?

(d) The diagram below shows the graph of the gradient function of a curve. For what value(s) of x does $f(x)$ have a local maximum? Justify your answer. [2]



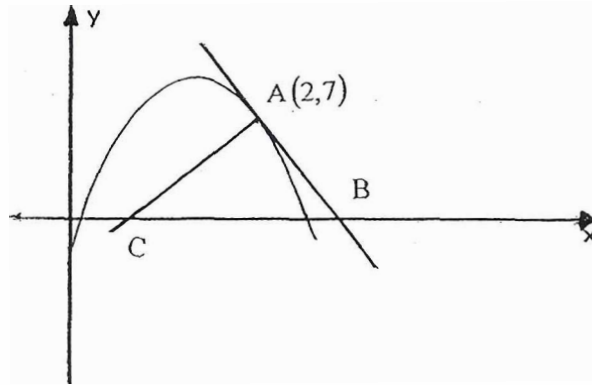
Question 6 (15 marks)

- (a) The diagram below shows the graphs of the functions $y = -x^2 + 7x - 6$ and $y = x + 2$.

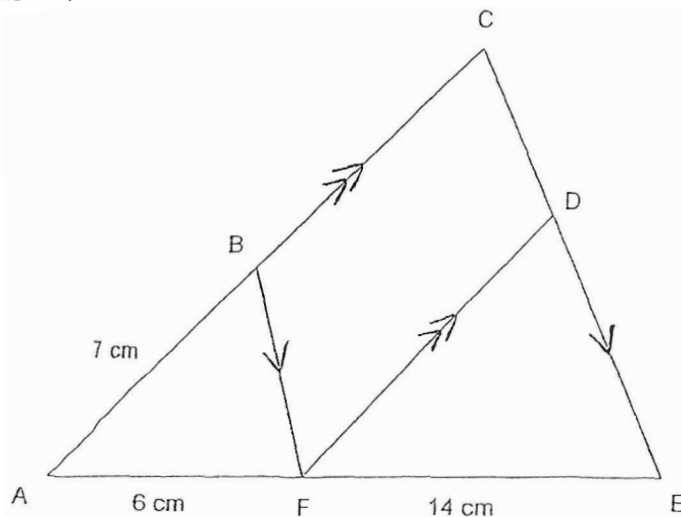


- (i) Show that the value of A and B is 2 and 4 respectively. [2]
- (ii) Calculate the area of the shaded region. [2]
- (iii) Write down a pair of inequalities that specify the shaded region. [1]

(b)



- (i) Find the equation of the tangent and the equation of the normal to the curve $y = -2x^2 + 6x + 3$ at the point $A(2,7)$. [4]
- (ii) The tangent cuts the x axis at B. The normal cuts the x axis at C as shown in the diagram. Find the values of B and C. Hence, find the area of $\triangle ABC$. [2]
- (c) In the diagram below AC is parallel to FD and BF is parallel to CE. B lies on AC, D lies on CE and F lies on AE. $AF = 6\text{cm}$, $FE = 14\text{cm}$ and $AB = 7\text{cm}$.



- (i) Find BC. [2]
- (ii) Find the ratio of BF to DE. [2]

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Q1

$$80 \times \frac{\pi}{180} = \frac{4\pi}{9}^{\circ}$$

Q1

2006

2U

TASK 2

Q2

$$\frac{17\pi}{12} = 255^{\circ}$$

Q3

$$i) \frac{d}{dx}(4x^2+5) = 8x$$

$$ii) \frac{d}{dx}((2x^3-5)^4) = 4 \times 6x^2 \times (2x^3-5)^3 \\ = 24x^2(2x^3-5)^3$$

$$iii) \frac{d}{dx} \left(\frac{3x+1}{2x-1} \right) = \frac{vu' - uv'}{v^2} \\ = \frac{(2x-1)(3) - (3x+1)(2)}{(2x-1)^2} \\ = \frac{\cancel{6x} - 3 - \cancel{6x} - 2}{(2x-1)^2} \\ = \frac{-5}{(2x-1)^2}$$

~~~~~

Q1 (c) (iv).

$$\begin{aligned}\frac{d}{dx} \left( x \sqrt{1-2x} \right) &= v u' + u v' \\&= \sqrt{1-2x} + x \left( \frac{1}{2} (1-2x)^{-\frac{1}{2}} (-2) \right) \\&= \sqrt{1-2x} - \frac{x}{\sqrt{1-2x}} \\&= \frac{1-2x-x}{\sqrt{1-2x}} \\&= \frac{1-3x}{\sqrt{1-2x}}.\end{aligned}$$

$$\begin{aligned}\text{d) i) } \int \frac{2}{x^4} \cdot dx &= \int 2x^{-4} \cdot dx \\&= \frac{2}{-3} x^{-3} + C \\&= -\frac{2}{3x^3} + C.\end{aligned}$$

$$\text{Q1) d) ii) } \int \sqrt{x} \, dx = \int x^{\frac{1}{2}} \, dx.$$

$$= \frac{2}{3} x^{\frac{3}{2}} + C$$

e)

~~$$f(x) = x^5.$$~~

~~$$f'(x) = \frac{x^6}{6}$$~~

~~$$f''(x) = \frac{x^7}{7 \times 6}$$~~
~~$$= \frac{x^7}{42}.$$~~

~~$$f''(2) = \frac{2^7}{42}$$~~

~~$$= 3.05 \text{ (2 dp)}$$~~

$$f(x) = x^5,$$

$$f'(x) = 5x^4$$

$$f''(x) = 20x^3$$

$$f''(2) = 20(2)^3$$

$$= 160$$

## QUESTION(2)

a)  $A(-1, 5) B(2, 1).$

Two pt.

~~$m_1 = \frac{5-1}{-1-2}$~~   
 ~~$= \frac{4}{-3}$~~

$$\frac{y-1}{x-2} = \frac{5-1}{-1-2}$$

$$-3(y-1) = 4(x-2)$$

$$-3y+3 = 4x-8$$

~~$4x-3y$~~   $y = \frac{4x-11}{-3}$

$C(4, k)$

$$y = \frac{4(4)-11}{-3}$$

$$= \frac{-3}{-3} + \frac{5}{-3} = \boxed{\frac{5}{3}}$$

$$= 1.$$

$\therefore k$  is 1 and  $C(4, \frac{5}{3})$ .

## Question 2.

$$b). \int \frac{x^2+1}{x^2} \cdot dx.$$

$$= \int 1 + \frac{1}{x^2} \cdot dx.$$

$$= \int 1 + x^{-2} \cdot dx.$$

$$= x + \frac{x^{-1}}{-1} \cdot dx.$$

$$= x - \frac{1}{x} + C.$$

$$c). \int_{-1}^1 (x-1)(x+1) \cdot dx.$$

$$= \int_{-1}^1 x^2 - 1 \cdot dx.$$

$$= \left[ \frac{x^3}{3} - x \right]_{-1}^1$$

$$= \left[ \frac{1}{3} - 1 \right] - \left[ -\frac{1}{3} + 1 \right].$$

$$= -\frac{2}{3} - \frac{2}{3}.$$

$$= -\frac{4}{3} = \underline{\underline{-1\frac{1}{3}}}$$



# QUESTION (2) (d).

i)

|   | 1 | 2 | 3 | 4  | 5  | 6  |
|---|---|---|---|----|----|----|
| 1 | 2 | 3 | 4 | 5  | 6  | 7  |
| 2 | 3 | 4 | 5 | 6  | 7  | 8  |
| 3 | 4 | 5 | 6 | 7  | 8  | 9  |
| 4 | 5 | 6 | 7 | 8  | 9  | 10 |
| 5 | 6 | 7 | 8 | 9  | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |

$$P(\text{sum} > 8) = \frac{10}{36} = \frac{5}{18}.$$

$$P(1,4 \text{ \& sum} > 8) = \frac{4}{36} = \frac{1}{9}.$$

(e) A(1,3) B(8,2) C(4,-1).

$$\text{Midpoint AB} = \frac{E}{2} = \left(\frac{9}{2}, \frac{5}{2}\right)$$

$$\text{Midpoint AC} = D = \left(\frac{4+1}{2}, \frac{3-1}{2}\right) = \left(\frac{5}{2}, 1\right).$$

$$\text{grad DE} = \frac{5/2 - 1}{9/2 - 5/2} = \frac{3/2}{2} = \frac{3}{4}.$$

$$\text{grad CB} = \frac{2 - -1}{8 - 4} = \frac{3}{4}.$$

CB // DE.

### Question 3

a) If  $y = (x^2 + 4)(x - 3)$ , solve  $\frac{dy}{dx} = 4$ .

$$y = x^3 - 3x^2 + 4x - 12$$

$$\frac{dy}{dx} = 3x^2 - 6x + 4$$

$$\therefore 4 = 3x^2 - 6x + 4$$

$$0 = 3x^2 - 6x$$

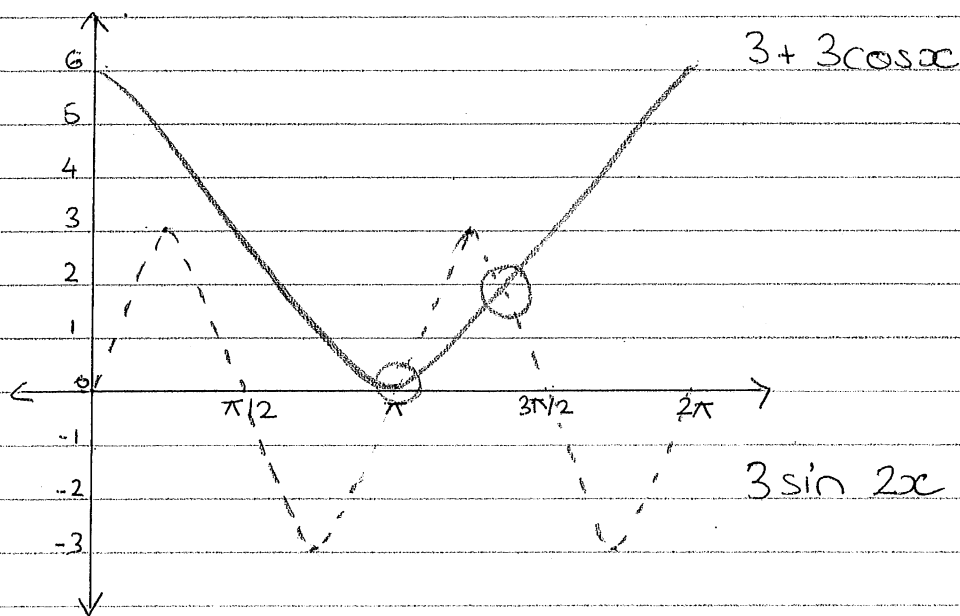
$$0 = 3x(x - 2)$$

$$3x = 0, \quad x - 2 = 0$$

$$x = 0, \quad x = 2$$

b) i)

iii)

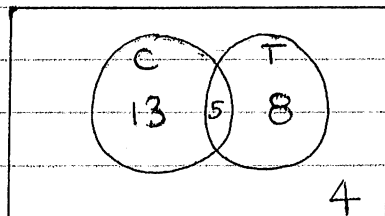


ii)  $y = 3\sin 2x$ . amplitude = 3

$$\text{period} = \frac{2\pi}{2} = \pi$$

iv) There are 2 solutions.

c) class.



$$P(\text{neither}) = \frac{4}{30} \\ = \frac{2}{15}$$

d)

|        |    |    |    |    |    |
|--------|----|----|----|----|----|
| $x$    | 0  | 10 | 20 | 30 | 40 |
| $f(x)$ | 20 | 14 | 12 | 7  | 8  |

$$h = \frac{40-0}{4} = \frac{40}{4} = 10$$

$$\int_0^{40} f(x) dx \doteq \frac{10}{3} [(20+8) + 4(14+7) + 2(12)] \\ \doteq \frac{10}{3} [28 + 4(21) + 2(12)] \\ = 453 \frac{1}{3} \text{ m}^2$$

e)  $f'(x) = 3x^2 - 1$  P.t. point (4, 1)

$$f(x) = \int f'(x) \\ = \int 3x^2 - 1 \\ = x^3 - x + C$$

$$f(4) = 1 \quad \therefore \quad 1 = 4^3 - 4 + C \\ 1 = 64 - 4 + C \\ 1 = 60 + C \\ C = -59$$

$$\therefore f(x) = x^3 - x - 59$$

### Question 4

a) i)  $P(\text{1st prize}) = \frac{5}{100}$   
 $= \frac{1}{20}$

ii)  $P(\text{2nd prize}) = P(\text{1st \& 2nd prize}) + P(\text{not 1st \& 2nd prize})$   
 $= (\frac{5}{100} \times \frac{4}{99}) + (\frac{95}{100} \times \frac{5}{99})$   
 $= \frac{1}{20}$

b)  $y = x^4 - 4x^3$   
 $y' = 4x^3 - 12x^2$

$$y'' = 12x^2 - 24x$$

i) SP @  $y' = 0$   
 $0 = 4x^3 - 12x^2$   
 $0 = 4x^2(x - 3)$   
 $4x^2 = 0, x - 3 = 0$   
 $x = 0, x = 3$

when  $x = 0, y = 0$   
 $\therefore$  SP @  $(0, 0)$   
when  $x = 3, y = 3^4 - 4 \times 3^3$   
 $= -27$   
 $\therefore$  SP @  $(3, -27)$

ii)  $y'' = 0$  helps determine nature of st. pts.

$$y''(0) = 0 - 0 = 0$$

$$y''(3) = 12 \times 3^2 - 24 \times 3 = 36$$

- possible pt of inflexion  $\therefore (3, -27)$  is a local minimum

check  $y'$  for sign of derivative.

|      |     |   |    |
|------|-----|---|----|
| $x$  | -1  | 0 | 1  |
| $y'$ | -16 | 0 | -8 |

$\therefore (0, 0)$  is a pt of inflexion.

iii)  $y'' > 0$  for concave up;  $\therefore 12x^2 - 24x > 0$   
 $x^2 - 2x > 0$   
 $x(x - 2) > 0$   
 $\therefore x > 2, x < 0$   
 $\therefore$  the curve is concave up for  $x > 2, x < 0$ .

iv)  $f'(x) < 0$  for curve decreasing.

$$4x^3 - 12x^2 < 0$$

$$x^2(x - 3) < 0$$

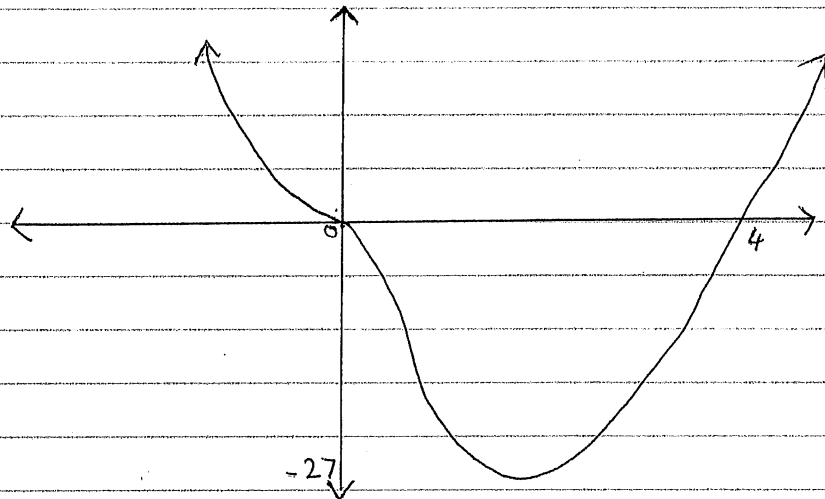
$$x^2 < 0, x < 3$$

$\therefore$  the curve is decreasing for  $x < 3, x \neq 0$

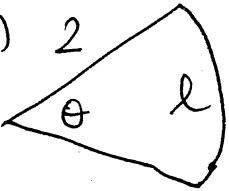
$$c) P(E) = 1/x$$

$$\begin{aligned} P(\text{not } E) &= 1 - 1/x \\ &= \frac{x-1}{x} \end{aligned}$$

Q4b) v)



Question (5). (a)

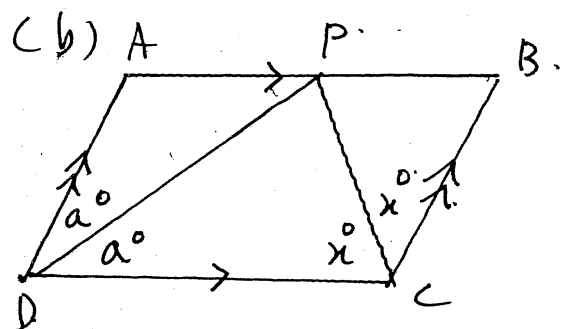
(i)   $A = \frac{1}{2} r^2 \theta$   
 $1.6 = 2\theta$   
 $\therefore \theta = 0.8$  \*

[2] i.e.  $0.8^\circ = \frac{180}{\pi} \times 0.8$   
 $\doteq 45^\circ 50'$  \*

(ii)  $l = r\theta$   
 $= 2 \times 0.8$   
 $= 1.6$  \*

$\therefore p = l + 2r = 5.6 \text{ m}$   
 $= 560 \text{ cm.}$  \*

[2]



## Solution to Section (c)

(i) In parm. ABCD.  
 $\angle ADC + \angle BCD = 180^\circ$   
 (co-interior  $\angle$ s are supplementary,  $AD \parallel BC$ )

$\therefore 2a + 2x = 180$

$a + x = 90^\circ$

$\angle DPC = 180^\circ - (a + x)$

(Angle sum of  $\triangle DPC$ )

i.e.  $\angle DPC = 180^\circ - 90^\circ$   
 $= 90^\circ$  [2]

(ii) In  $\triangle APD$ . [2]

$\angle PDC = \angle APD = a^\circ$

(alternate  $\angle$ s,  $AP \parallel DC$ )

$\therefore \angle ADP = \angle APD = a^\circ$

i.e.  $\triangle ADP$  is isosceles.

(iii) Similar (from (ii))

•  $BP = BC$ . ( $\triangle BPC$  is isosceles)  
 and  $AD = AP$  (proven).

• but  $AD = BC$   
 (opposite sides of a parm equal). [2]

$AB$   
 $\therefore = AP + PB$   
 $= AD + BC$   
 $= 2BC$ .

(c)

(i)  $\int_0^8 f(x) dx$

$= \frac{1}{4}(\pi \times 2^2) + 2^2 + \frac{1}{2}(2) \times 2$   
 $- \frac{1}{2}(2) \times 2$  [2]

$= \pi + 4$  (7.142)

(ii) At  $x = 4$   $f$  is [1]  
 not differentiable.

(cd) At  $x = 5$  [2]

$f'(5) = 0$

$f'(5-\epsilon) > 0$  (for small positive  $\epsilon$ )

$f'(5+\epsilon) < 0$ .

$\therefore$  By the 1<sup>st</sup> derivative test  $f(x)$  has a local maximum at  $(5, 0)$ .

# Question (6)

(a)  $y = -x^2 + 7x - 6$   
 $y = x + 2$

$$-x^2 + 7x - 6 = x + 2$$

(i)  $x^2 - 6x + 8 = 0$

$$(x - 4)(x - 2) = 0$$

$$\therefore x = 2, y = 4$$

$$x = 4, y = 6$$

$$(2, 4) \quad (4, 6) \quad [2]$$

$\therefore A = 2$  and  $B = 4$ .

(ii)  $A = \int_2^4 [(-x^2 + 7x - 6) - x - 2] dx$

$$= \int_2^4 (-x^2 + 6x - 8) dx$$

$$= \left[ -\frac{x^3}{3} + 3x^2 - 8x \right]_2^4$$

$$= \left[ -\frac{64}{3} + 48 - 32 \right] - \left[ -\frac{8}{3} + 12 - 16 \right]$$

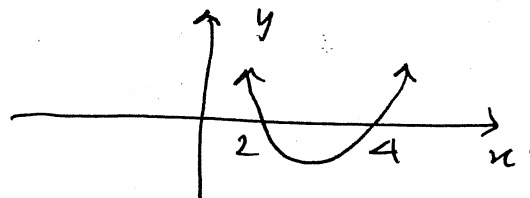
$$= -\frac{56}{3} + 16 + 4$$

$$= \frac{4}{3} \quad [2]$$

(iii)  $-x^2 + 6x - 8 > 0$

$$x^2 - 6x + 8 < 0$$

$$(x - 4)(x - 2) < 0$$



$$2 < x < 4 \quad [2]$$

(b)  $y = -2x^2 + 6x + 3$

$A(2, 7)$

$$\frac{dy}{dx} = -4x + 6$$

$$\therefore \frac{dy}{dx} \Big|_{x=2} = -2$$

$$\therefore y - 7 = -2(x - 2)$$

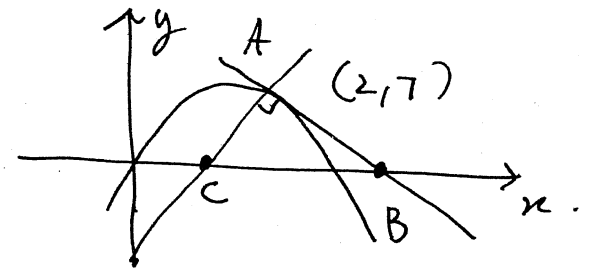
$$\therefore 2x + y - 11 = 0 \quad [+g+]$$

$$\therefore y - 7 = \frac{1}{2}(x - 2)$$

$$\therefore 2y - 14 = x - 2$$

$$\therefore x + 2y + 12 = 0 \quad [4]$$

(normal)



$$2x = 11 \quad x = \frac{11}{2} \quad [2]$$

$$x = -12$$

$$B\left(\frac{11}{2}, 0\right) \quad C(-12, 0)$$

$$\therefore \text{Area} = \frac{1}{2} \times 17\frac{1}{2} \times 7$$

(c) Let  $BC = x$ .

(i)  $\frac{7}{7+x} = \frac{6}{10}$

$$70 = 21 + 3x$$

$$9 = 3x$$

$$x = 49/3$$

(ii)  $BF : DE$

$$7 : \frac{49}{3}$$

$$21 : 49$$

$$3 : 7$$

$$[2]$$